



Improvement of Cycling Performance of FeF₃-Based Lithium-Ion Battery by Boron-Based Additives

Kiyoshi Kumagae,^a Ken-ichi Okazaki,^z Keitaro Matsui, Hideyuki Horino, Toshiro Hirai,* Jun-ichi Yamaki,* and Zempachi Ogumi**

Office of Society-Academia Collaboration for Innovation, Kyoto University, Gokasho, Uji, Kyoto 611-0011, Japan

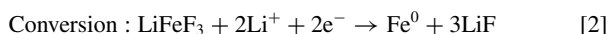
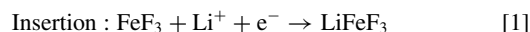
Two boron-based additives, tris(pentafluorophenyl)borane (TPFPB) and tris(1H,1H-heptafluorobutyl)borate (THFBuBO), are studied for their effects on the cycling performance of the FeF₃/Li battery. When dissolved in the electrolyte consisting of LiPF₆-ethylene carbonate/1,2-dimethyl carbonate, these compounds act as F[−] receptors to improve the electrochemical properties of the FeF₃ cathode and cycling performance. The initial discharge capacity and voltage profile of FeF₃/Li battery displayed little change with the additives. However, TPFPB and THFBuBO improved the capacity retention, with respective retention rates of 85.2% and 86.5% at the 10th cycle, while this value was 68.6% when no additive was used. Inductively coupled plasma atomic emission spectroscopy analysis showed that LiF was dissociated by TPFPB or THFBuBO in the electrolyte, and its solubility was higher with TPFPB. THFBuBO has the weak affinity for F[−] ions to allow reversible formation/dissociation of the [THFBuBO-F][−] complex. Therefore, THFBuBO maintained the capacity of the FeF₃/Li battery better than TPFPB.

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The lithium ion battery (LIB), with its superior cycling performance and moderately high energy density, has been widely used for mobile devices and greatly contributed to their prevalence.^{1,2} The currently available commercial LIBs are based on the intercalation/deintercalation reactions of Li⁺ ions with the cathode material (e.g. LiCoO₂ and LiFePO₄), and the anode material (e.g. graphite). During the reversible intercalation/deintercalation reaction, the volume change in the crystal lattice of cathode materials is quite small. This structural stability contributes to the extremely high cycle stability, high rate capability, and long lifetime of LIB. However, the number of the Li⁺ ions that can be inserted per site is intrinsically limited by the electrode material. Meanwhile, impending larger devices such as electric vehicles, drones, wearing-type assist-robots require ever-higher energy density. Therefore, cathode active materials with larger capacities are being developed to increase the energy density of LIB. Cathode materials based on the conversion reactions have been proposed,^{3,4} which possess theoretical capacities higher than that of intercalation-type materials. Sulfur⁵ (theoretical capacity 1675 mAh g^{−1}) and metal halides^{6–8} (MF_x and MCl_y, M = Bi, Co, Cu, Fe, Ni, etc.) have been researched extensively for this purpose.

Metal fluoride is one of the more promising candidates, having high energy density³ and high stability. Iron trifluoride (FeF₃) has attracted considerable attention as an active material, because of its low cost, non-toxicity, and a high theoretical capacity (712 mAh g^{−1} for three-electron Fe³⁺ → Fe⁰ reaction).⁶ The electrochemical performance of FeF₃ in the cathode of LIB was first reported by Arai and co-workers.⁹ The discharge reaction of FeF₃ in LIB can be divided into two steps:



Reaction 1 is the insertion of Li⁺ ion involving the reduction of Fe³⁺ to Fe²⁺, and Reaction 2 is the conversion involving the reduction of Fe²⁺ to Fe⁰. However, FeF₃ suffers from low electron conductivity and low reaction reversibility. Since the low conductivity of FeF₃ hinders the full utilization of its high capacity, many researchers have examined ways to improve its electrochemical properties, such as using nanostructured-FeF₃ and/or FeF₃ nanocomposites.^{10–12} For example, Badway and co-workers achieved a discharge capacity close to the theoretical limit by using a ball-milled FeF₃/C nanocomposite electrode.¹¹ Similarly, Li and co-workers have synthesized FeF₃

nanocrystals chemically and acquired ~800 mAh g^{−1} in the voltage range of 4.5–1.0 V using FeF₃/C composites.¹² Other studies to improve the low conductivity of FeF₃ have been reported as well.^{13–16}

The low reversibility of FeF₃ redox reactions is another problem, especially the conversion reaction, although the detailed mechanism leading to deteriorated cycling remains unclear.^{17–19} In the CuF₂/C composite cathode, Wang and co-workers reported that the converted Cu segregates into large particles during discharge process, thereby reducing the reaction reversibility.¹⁹ We suspect that in FeF₃/C composite cathode, similar segregation of Fe and LiF formed in the conversion reaction during discharge is one of the factors of low reversibility. Part of the segregated LiF would not participate in the reconversion reaction during the next charge process, causing lowered cycling performance.

The above results suggest that an additive in the electrolyte that helps dissociating the inactive LiF can enhance cycling performance. Boron-based compounds (e.g. borane- and borate-compounds) have been studied extensively as anion receptors (ARs) that can be complexed with fluoride anions (F[−]) in solutions.^{20–24} The reaction between LiF and the AR is represented as the following.



Reaction 3 produces Li⁺ ion and the fluorinated-AR complex ion [AR-F][−] in solution. Boron-based ARs have been proposed for LiF-containing electrolytes for the intercalation-based LIB, by dissolving LiF according to Reaction 3.^{21–23} The ARs have also been used to remove LiF from the passivation film on the cathode surface in the intercalation-based LIB in order to improve the battery cycle life.^{24,25} Here we hypothesize that cycling performance in the conversion-based FeF₃/Li battery can be enhanced in a similar manner, if the formed [AR-F][−] can be reused in the reconversion reaction during the following charge process. As far as we know, such attempts have not been reported. In this study, boron-based ARs with different levels of affinity for F[−] were used, and the solubility of LiF and the conductivity of LiF electrolyte solution containing the ARs were evaluated. FeF₃-based coin-type LIBs containing the ARs were also fabricated to investigate the effects of these additives on the cycling performance.

Experimental

The FeF₃/C composite electrode was prepared as follows. Anhydrous iron trifluoride (FeF₃) powder was purchased from Rare Metallic Co., Ltd. and used without further treatment. The anhydrous FeF₃ powder was mixed with acetylene black (DENKA BLACK, Denka Company Limited) in a weight ratio of 70:25. The powder mixture was put into a stainless steel container involved zirconia grinding bowl under argon atmosphere, and then it was ball-milled at 200 rpm

*Electrochemical Society Member.

**Electrochemical Society Fellow.

^aPresent address: Advanced Materials Laboratories, Sony Corporation, Atsugi 243-0014, Japan.

^zE-mail: k-okazaki@saci.kyoto-u.ac.jp

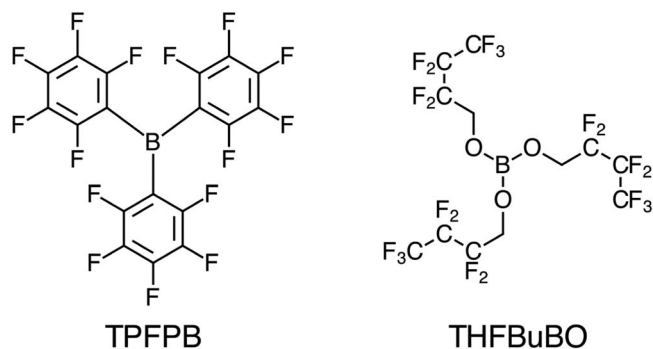


Figure 1. Chemical structures of boron-based additives TPFPB and THF-BuBO.

for 24 h with a planetary ball mill (Premium line P-7, Fritsch). An aliquot of 0.95 g of thus prepared FeF₃/C composite powder, 0.05 g of polyvinylidene difluoride (HSV900, Arkema Inc.) binder and 4.88 g of N-methyl-2-pyrrolidone (Kishida Chemical Co., Ltd., LBG grade) were weighed exactly in a perfluoroalkoxy-Teflon bottle, and then mixed for 12 h with a shaker to prepare a slurry. The prepared slurry was cast with a baker-type applicator (Hohsen Corp.) on a piece of aluminum foil as a current collector. The resulting FeF₃/C composite electrode was dried at 363 K for at least one night under vacuum.

LiPF₆, ethylene carbonate (EC), and 1,2-dimethylene carbonate (DMC) of LBG grade were purchased from Kishida Chemical. The electrolyte was prepared by dissolving 1.0 mol dm⁻³ LiPF₆ in a mixture of EC and DMC (1:1 v/v). Tris(pentafluorophenyl) borane (TPFPB, Aldrich) and tris(1H,1H-heptafluorobutyl) borate (THF-BuBO, Synquest Laboratories) were used as anion receptors (ARs) in this study, because they are well known for promoting the dissociation of LiF in organic solvents.^{20,22} Their purities were more than 95% and they were used as received without further purification. The chemical structures of the ARs are shown in Figure 1. The solubilities of these ARs in the electrolyte were different each other; THFBuBO was difficult to dissolve in comparison with TPFPB. In this study, therefore, nearly saturated amounts of each AR were added in the electrolyte. The concentrations of the additives in the electrolyte were 0.1 mol dm⁻³ for TPFPB and 0.02 mol dm⁻³ for THFBuBO.

The discharge and charge profiles of the FeF₃/Li battery were measured using the standard CR2032 coin-type cell configuration. The cathode disk was fabricated by punching from the FeF₃/C composite electrode, with a projected area of 1.77 cm². Each cell consisted of a cathode, a lithium anode (2.01 cm²) and 1.0 mol dm⁻³ LiPF₆ in EC-DMC electrolyte with or without AR, and the cell was assembled in an argon-filled glove box. The assembled cells were placed in a thermostatic oven at 298 K. Then the discharge-charge tests were carried out at the charging rate of 0.05 C in the voltage range of 1.0–4.5 V using a charge/discharge system (HJ1001SD8, Hokuto Denko Corp.).

The fluoride ion affinity of the ARs was evaluated by measuring the ionic conductivity of EC-DMC solutions with saturated LiF (Wako Chemicals), and none or one of the additives. The test solutions were prepared in an argon-filled glove box as follows. LiF (0.05 mol dm^{-3}) was dispersed in EC-DMC alone or with 0.05 mol dm^{-3} additive, and then the dispersion was stirred at 318 K overnight. Saturation of LiF was confirmed by observing solid LiF at the bottom of the vial after stirring. The supernatant with saturated LiF was used for ionic conductivity testing. The ionic conductivity measurements were carried out at 298 K in the frequency range from 100 kHz to 1 Hz with a Solartron SI1287 electrochemical interface, together with a Solartron 1255B frequency response analyzer. An electrochemical cell with stainless steel electrodes (area: 0.785 cm^2 , distance: 0.5, 1.25 and 2.0 mm) was used for these measurements.

The solubilities of LiF, FeF₃, and iron difluoride (FeF₂, Rare Metallic Co., Ltd.) in the EC-DMC solution with ARs were measured by

Table I. Ionic conductivities of the LiF/EC-DMC electrolyte with and without boron-based ARs at 298 K.

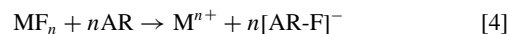
	TPFPB + LiF	THFBuBO + LiF	only LiF
Conductivity (S cm ⁻¹)	1.13 × 10 ⁻³	5.40 × 10 ⁻⁵	7.33 × 10 ⁻⁷

inductively coupled plasma atomic emission spectroscopy (ICP-AES; SPS3520UV, SII NanoTechnology Inc.). The test solutions were prepared as follows. To each electrolyte, 0.1 mol dm⁻³ AR and a fluoride salt equivalent to 0.3 mol dm⁻³ F⁻ ions (i.e. 0.1 mol dm⁻³ FeF₃, 0.15 mol dm⁻³ FeF₂ or 0.3 mol dm⁻³ LiF) were co-dispersed in the EC-DMC in argon atmosphere, and the resulting suspension stirred for one week at 298 K. It was then centrifuged at 30,000 G (~18,800 rpm) for 60 min at 298 K, and the transparent supernatant was used for testing. The solubilities were estimated from the concentrations of dissolved Fe and Li measured by ICP-AES.

Results and Discussion

In order to evaluate the fluoride-ion affinities of the two boron-based additives TFPFB and THFBuBO, the ionic conductivities of the corresponding LiF/EC-DMC solutions were measured. The results obtained with and without additives at 298 K are listed in Table I. Without the additive the conductivity was very low (7.33×10^{-7} S cm^{-1}), because the LiF salt is insoluble in EC-DMC. In contrast, the addition of THFBuBO or TFPFB increased the conductivity to the order of 10^{-5} and 10^{-3} S cm^{-1} , respectively. This increase suggests that LiF does react with AR according to Reaction 3, because the resulting Li^+ and the $[\text{AR-F}]^-$ complex ions contribute to the ionic conduction. The significantly higher improvement in conductivity by TFPFB over THFBuBO is attributed to the strength of the Lewis acid at the boron center. According to the molecular structures shown in Fig. 1, the pentafluorophenyl group in TFPFB has much stronger electron-withdrawing effect than the $\text{CF}_3\text{CF}_2\text{CF}_2\text{CH}_2\text{O}$ group in THFBuBO.²⁰ Therefore, TFPFB is a stronger Lewis acid and it increased the solubility of LiF more than THFBuBO does.

The dissolution of LiF salt in EC-DMC with TFPBPB or THFBuBO was confirmed by ICP-AES analysis. The solubilities of the cathode active material (FeF_3) and the divalent fluoride FeF_2 as a comparison were also evaluated under the same condition, and the results are listed in Table II. Without the additives, the concentrations of each fluoride salt in EC-DMC were at very low to undetectable levels, implying that they do not dissolve in EC-DMC alone. In contrast, although the results of FeF_3 and FeF_2 are mentioned later, the solubility of LiF increased remarkably by the additives. Table II also lists the reaction ratios from the prepared AR (0.1 mol dm^{-3}) to the $[\text{AR-F}]^-$ ion in the following reaction



$$\text{Reaction ratio (\%)} : n \times C_M (\text{mol dm}^{-3}) / 0.1 (\text{mol dm}^{-3}) \times 100 \quad [5]$$

where M and n represent the metal atom (Fe or Li) and its valence ($n = 1-3$), respectively. C_M is the concentration of metal ion listed in Table II. By addition of TPFPB as the AR in the LiF/EC-DMC dispersion, almost all the LiF reacted with TPFPB via Reaction 4, and the concentration of Li^+ ion reached to $\sim 0.10 \text{ mol dm}^{-3}$. When THFBuBO was added as the AR, the reaction ratio of AR was only 1/20 that of TPFPB. This value corresponds well to the measured ion conductivities (with a ratio of 1/21). The difference of the solubility of LiF indicates that TPFPB has a stronger affinity for the fluoride ion than THFBuBO. In contrast, the AR reaction ratios for FeF_3 and FeF_2 were less than 2% with either TPFPB or THFBuBO. Therefore, the cathode active material is much harder to dissolve than LiF ($\text{LiF} \gg \text{FeF}_3$) in the presence of ARs.

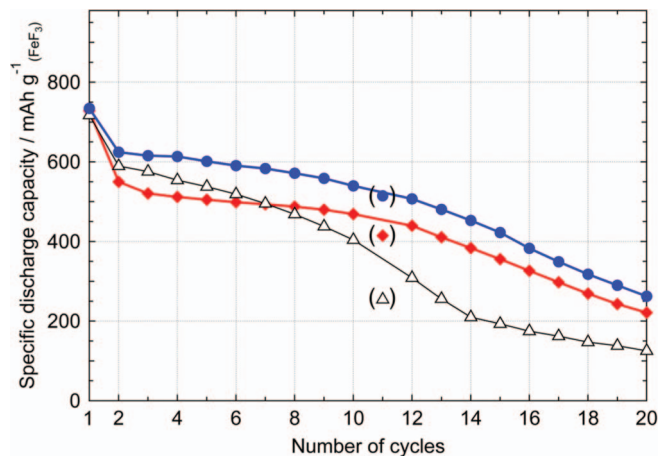
Figure 2 shows the discharge-charge profiles for the first cycle of the FeF₃/Li coin-type cells with and without additives. The cells were

Table II. Solubilities of LiF, FeF₃, and FeF₂ in EC-DMC (1:1) with and without boron-based ARs, from ICP-AES measurement. “DL” is an abbreviation of the detection limit. The corresponding reaction ratios of AR are also listed.

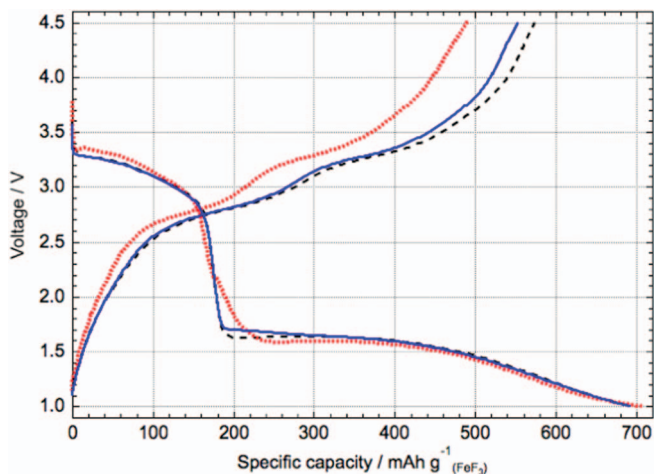
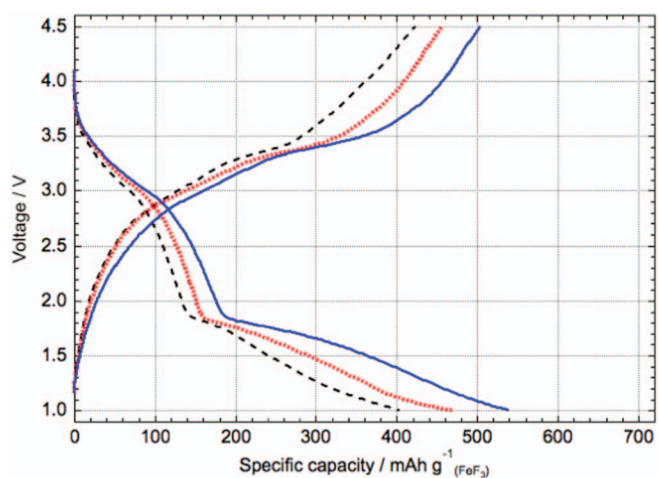
	Li(LiF)			Fe(FeF ₃)			Fe(FeF ₂)		
	salt only	with TFPFB	with THFBuBO	salt only	with TFPFB	with THFBuBO	salt only	with TFPFB	with THFBuBO
concentration ($\times 10^{-3}$ mol dm ⁻³)	<DL	100.9	5.3	<DL	0.63	0.29	<DL	0.95	0.27
reaction ratio of AR (%)	-	100.9	5.3	-	1.9	0.9	-	1.9	0.5

tested between 1.0 and 4.5 V at the rate of 0.05 C under a constant current condition at 298 K. In the discharge profile without additives (dashed line), a plateau at 3.3 V was observed, and then the voltage decreased gradually to ~2.8 V. A second plateau was observed from 1.6 V for ca. 500 mAh g⁻¹ in specific capacity based on the weight of FeF₃ in the composite electrode. The first plateau between 3.3 and 1.6 V is ascribed to Reaction 1 (the insertion of Li⁺ ion and reduction of Fe³⁺ to Fe²⁺), and the second between 1.6 and 1.0 V is due to Reaction 2 (the conversion reaction involving the reduction of Fe²⁺ to Fe⁰). The reported theoretical capacities of these two parts are 237 and 474 mAh g⁻¹, respectively.²⁶ In our result, the capacity of the insertion reaction was slightly smaller than the theoretical value, while the total discharge capacity was very close to the theoretical one. In contrast, a slight increase in discharge capacity was observed between 2.2 and 1.6 V in the profile with TFPFB. When the cell was disassembled after the initial discharge, the surface of lithium metal electrode had been discolored. On the other hand, the increase of capacity and the color change of lithium electrode were rarely observed by use of THFBuBO. Therefore, these results suggest that TFPFB would react at the surface of lithium metal in the initial discharge. In the subsequent charge profiles, the capacities were lower than the corresponding initial discharge capacities. When TFPFB was added to the electrolyte, the charge capacity decreased to ca. 85% of the value without additive. The difference among these charge curves was mainly in the conversion region. This decrease of the charge capacity by TFPFB will be discussed later.

The specific discharge capacities for the FeF₃/Li cells with and without ARs to 20 cycles are plotted in Figure 3. The specific capacity was estimated from the weight of FeF₃ in the composite electrode. The discharge capacities of all three cells were well maintained at the 10th cycle, compared with the previous reports using until the conversion region.^{13,14,16} After the 10th charge, these cells were rested for one week; therefore, the consequent discharge capacities at 11th

**Figure 3.** Specific discharge capacity versus cycle number for FeF₃/Li coin-type cell with 1.0 mol dm⁻³ LiPF₆/EC-DMC electrolyte, in the voltage range of 4.5–1.0 V and at the rate of 0.05 C at 298 K. Solid circles: with THFBuBO, solid diamonds: with TFPFB, and open triangles: without additives. After the 10th cycle, the cells were rested for one week.

cycle decreased temporarily. This decrease of capacities will discuss in detail elsewhere. In all cases, there is an obvious decrease of discharge capacity between the first and second cycles. The difference would be attributed to the changes of spatial arrangement and composition of reactants during the initial discharge-charge process shown in Fig. 2. From the third to the 10th cycle, the shapes of the discharge-charge were similar to those at the second cycle, although the corresponding capacities gradually decreased. The 10th discharge-charge profiles with and without ARs are shown in Figure 4. Compared to the initial cycle (Fig. 2), no significant changes such as side reactions with a large

**Figure 2.** Initial discharge-charge profiles of FeF₃/Li coin-type cell with 1.0 mol dm⁻³ LiPF₆/EC-DMC electrolyte in the voltage range of 1.0–4.5 V, at the rate of 0.05 C at 298 K. Solid lines: with THFBuBO, dotted lines: with TFPFB, and dashed lines: without additives.**Figure 4.** Discharge-charge profiles of FeF₃/Li coin-type cell with 1.0 mol dm⁻³ LiPF₆/EC-DMC electrolyte at the 10th cycle. Solid lines: with THFBuBO, dotted lines: with TFPFB, and dashed lines: without additives.

capacity were observed in the discharge-charge profiles. Therefore, we discuss the cycling performance of these FeF₃/Li cells based on the capacities at the second cycle.

At the second cycle, the discharge capacities of cells with THFBuBO, TFPFB or without ARs were 624, 550, and 589 mAh g⁻¹, respectively. The discharge retention ratio is defined as the ratio between the discharge capacities at the 10th and the second cycles. Without ARs, the discharge capacity at the 10th cycle was 404 mAh g⁻¹, corresponding to a retention ratio of 68.6%. In contrast, the retention ratios with THFBuBO and TFPFB were estimated to 86.5 and 85.2%, respectively. That is, the discharge capacity retention was enhanced by the addition of either TFPFB or THFBuBO. These results clearly indicate that the ARs can significantly improve the cycling performance of the FeF₃/Li cell. In addition, the discharge capacity without ARs was dropped to ca. 100 mAh g⁻¹ at the 20th cycle. The active material might have been exfoliated or detached by the expansion/contraction during discharge-charge cycle between 1.0 and 4.5 V. Therefore, it should be improved the manufacturing of the FeF₃/C composite electrode.

Now we consider one of the causes of the decreasing capacity during cycling. Since the charge reaction occurs at the solid-solid interface between Fe and LiF, participating molecules would have to be located close to each other in the composite electrode. As the cycling progresses, Fe and LiF are gradually segregated, and the portions of Fe and LiF in the segregated domains no longer contribute to the charge and discharge reactions. In our result shown in Fig. 3, it is considered that the discharge product LiF was partially dissolved into the electrolyte by ARs via Reaction 3. Accordingly, a new reaction involving Fe and [AR-F]⁻ would occur at the solid-liquid interface. In other words, it is considered that part of the inactive LiF dissociated by ARs can react with Fe again in the reconversion reaction. Previous papers have reported that the fluorinated-AR can release its fluoride ion reversibly.^{27,28} The improvement of cycling performance can be explained by the reversibility of the fluorinated-ARs. The bonding strength between AR and F⁻ ion (fluoride ion affinity) is particularly important for this reversibility, so that the fluoride anion extracted from the inactive LiF by AR would be available to react with Fe in the charge process again.

The solubility results have shown that TFPFB has a higher fluoride ion affinity than THFBuBO. In Fig. 3, the discharge capacity with TFPFB decreased remarkably between the first and second cycles. This depression of the second discharge capacity is caused by the depression of charge capacity in the first cycle: although TFPFB dissociated LiF according to Reaction 3 at the first discharge, its strong fluoride ion affinity makes it harder for the fluoride ions to be released from [TFPFB-F]⁻.²⁷ The reaction between [TFPFB-F]⁻ and Fe in the charge process was therefore more difficult than that of THFBuBO, leading to significantly decreased second discharge capacity. These results indicate that the optimal cycling performance of the FeF₃/Li battery is attained by adding an AR with the appropriate level of fluoride ion affinity, such as THFBuBO. This concept could apply to other conversion-based LIBs using metal fluorides as the cathode active material. In order to find ARs with the optimal fluoride ion affinity, we are currently working on computational and experimental methods.

Conclusions

The cycle performance of the FeF₃-based LIB was improved by the addition of boron-based ARs (THFBuBO and TFPFB) in the electrolyte. Excellent capacity retention was obtained with an AR (e.g. THFBuBO) possessing medium level of affinity for F⁻, rather than one with strong affinity (e.g. TFPFB).

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References

1. M. Armand and J. M. Tarascon, *Nature*, **451**, 652 (2008).
2. J. B. Goodenough and Y. Kim, *Chem. Mater.*, **22**, 587 (2010).
3. G. G. Amatucci and N. Pereira, *J. Fluorine Chem.*, **128**, 243 (2007).
4. J. Cabana, L. Monconduit, D. Larcher, and M. R. Palacin, *Adv. Mater.*, **22**, E170 (2010).
5. X. Ji, K. T. Lee, and L. F. Nazar, *Nat. Mater.*, **8**, 500 (2009).
6. H. Li, P. Balaya, and J. Maier, *J. Electrochem. Soc.*, **151**, A1878 (2004).
7. M. Bervas, F. Badway, L. C. Klein, and G. G. Amatucci, *Electrochem. Solid-State Lett.*, **8**, A179 (2005).
8. A. J. Gmitter, F. Badway, S. Rangan, R. A. Bartynski, A. Halajko, N. Pereira, and G. G. Amatucci, *J. Mater. Chem.*, **20**, 4149 (2010).
9. H. Arai, S. Okada, Y. Sakurai, and J. Yamaki, *J. Power Sources*, **68**, 716 (1997).
10. F. Badway, N. Pereira, F. Cosandey, and G. G. Amatucci, *J. Electrochem. Soc.*, **150**, A1209 (2003).
11. F. Badway, F. Cosandey, N. Pereira, and G. G. Amatucci, *J. Electrochem. Soc.*, **150**, A1318 (2003).
12. T. Li, L. Li, Y. L. Cao, X. P. Ai, and H. X. Yang, *J. Phys. Chem. C*, **114**, 3190 (2010).
13. L. S. Li, F. Meng, and S. Jin, *Nano Lett.*, **12**, 6030 (2012).
14. Q. Chu, Z. Xing, X. Ren, A. M. Asiri, A. O. Al-Youbi, K. A. Alamry, and X. Sun, *Electrochim. Acta*, **111**, 80 (2013).
15. Y. Shen, X. Wang, H. Hu, M. Jiang, X. Yang, and H. Shu, *J. Power Sources*, **283**, 204 (2015).
16. X. Fan, Y. Zhu, C. Luo, T. Gao, L. Suo, S.-C. Liou, K. Xu, and C. Wang, *J. Power Sources*, **307**, 435 (2016).
17. H. J. Tan, H. L. Smith, L. Kim, T. K. Harding, S. C. Jones, and B. Fultz, *J. Electrochem. Soc.*, **161**, A445 (2014).
18. M. Sina, R. Thorpe, S. Rangan, N. Pereira, R. A. Bartynski, G. G. Amatucci, and F. Cosandey, *J. Phys. Chem. C*, **119**, 9762 (2015).
19. F. Wang, R. Robert, N. A. Chernova, N. Pereira, F. Omenya, F. Badway, X. Hua, M. Ruotolo, R. Zhang, L. Wu, V. Volkov, D. Su, B. Key, M. S. Whittingham, C. P. Grey, G. G. Amatucci, Y. Zhu, and J. Graetz, *J. Am. Chem. Soc.*, **133**, 18828 (2011).
20. H. S. Lee, X. Q. Yang, C. L. Xiang, J. McBreen, and L. S. Choi, *J. Electrochem. Soc.*, **145**, 2813 (1998).
21. H. S. Lee, X. Sun, X. Q. Yang, and J. McBreen, *J. Electrochem. Soc.*, **149**, A1460 (2002).
22. X. Sun, H. S. Lee, X. Q. Yang, and J. McBreen, *J. Electrochem. Soc.*, **149**, A355 (2002).
23. L. F. Li, H. S. Lee, H. Li, X. Q. Yang, K. W. Yang, W. S. Yoon, J. McBreen, and X. J. Huang, *J. Power Sources*, **184**, 517 (2008).
24. Z. H. Chen and K. Amine, *J. Electrochem. Soc.*, **153**, A1221 (2006).
25. J. M. Zheng, J. Xiao, M. Gu, P. J. Zuo, C. M. Wang, and J. G. Zhang, *J. Power Sources*, **250**, 313 (2014).
26. N. Yamakawa, M. Jiang, B. Key, and C. P. Grey, *J. Am. Chem. Soc.*, **131**, 10525 (2009).
27. W. C. West, J. F. Whitacre, N. Leifer, S. Greenbaum, M. Smart, R. Bugga, M. Blanco, and S. R. Narayanan, *J. Electrochem. Soc.*, **154**, A929 (2007).
28. X. W. Cui, J. Chen, T. F. Wang, and W. X. Chen, *Sci. Rep.*, **4**, 5310 (2014).